

QUEBEC CITY BEAUPORT, QC, Canada

WMO: 715780

Lat: 46.8371N

Lon: 71.1973W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 99-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-20.5	-29.7	0.2	-22.6	-26.8	0.3	-20.0	13.3	-5.5	11.7	-6.7	4.9	240	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.9	21.1	28.2	20.1	26.5	19.3	23.1	27.6	21.8	26.1	20.7	24.6	5.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	16.2	25.6	20.3	15.0	24.2	19.2	14.0	23.1	68.4	27.9	63.7	26.2	59.9	24.6	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	10.6	9.2	-26.0	32.8	3.2	2.1	-28.4	34.3	-30.2	35.5	-32.1	36.6	-34.4	38.1
			-26.7	24.3	2.9	1.1	-28.8	25.1	-30.5	25.7	-32.1	26.3	-34.2	27.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-10.1	-8.1	-2.7	4.6	11.8	17.2	20.7	19.7	15.4	8.1	1.9	-5.8	(6)
(7)		DBStd	11.47	6.47	5.49	5.25	3.78	3.90	3.58	2.93	2.92	3.61	4.05	4.03	5.41	(7)
(8)		HDD10.0	2534	622	507	395	167	22	0	0	0	2	87	243	490	(8)
(9)		HDD18.3	4643	881	740	653	413	205	63	10	19	102	318	492	748	(9)
(10)		CDD10.0	1125	0	0	0	4	79	216	331	302	164	27	1	0	(10)
(11)		CDD18.3	193	0	0	0	0	4	29	82	62	14	1	0	0	(11)
(12)		CDH23.3	1332	0	0	0	2	71	223	551	378	102	6	0	0	(12)
(13)		CDH26.7	346	0	0	0	0	18	63	152	87	25	1	0	0	(13)
(14)	Wind	WSAvg	4.3	4.5	4.6	4.5	4.9	4.9	4.0	3.5	3.5	3.6	4.4	4.4	4.5	(14)
(15)	Precipitation	PrecAvg	1243	86	73	77	93	102	118	132	124	119	115	98	102	(15)
(16)		PrecMax	1419	153	147	148	221	261	238	259	210	232	244	165	206	(16)
(17)		PrecMin	981	32	23	22	32	45	45	44	28	78	48	27	50	(17)
(18)		PrecStd	123	33	28	30	49	43	40	47	38	32	45	31	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	4.5	11.8	20.5	29.2	31.4	33.2	31.1	29.7	23.3	13.6	6.2	(19)
(20)		MCWB	3.7	2.9	7.0	12.0	17.1	21.5	24.3	22.4	21.8	17.2	10.4	5.1	(20)	
(21)		2%	DB	2.5	2.4	7.7	16.1	25.2	28.5	30.0	29.1	26.3	19.5	10.9	3.9	(21)
(22)		MCWB	1.5	1.0	3.7	8.4	15.4	19.7	22.0	20.7	19.6	14.4	8.9	2.6	(22)	
(23)		5%	DB	1.1	1.1	5.2	12.6	21.5	25.9	28.2	27.2	23.4	16.3	8.7	2.6	(23)
(24)		MCWB	0.4	-0.1	2.6	6.8	14.2	18.0	20.7	19.9	17.9	13.1	6.8	1.5	(24)	
(25)		10%	DB	-1.1	-0.2	3.4	10.1	18.5	23.6	26.4	25.4	21.3	13.7	7.1	1.3	(25)
(26)		MCWB	-2.0	-1.1	1.3	5.7	12.3	17.0	19.8	19.2	16.9	10.9	5.6	0.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.2	8.1	13.7	20.0	22.8	25.2	23.7	22.6	18.1	11.6	5.6	(27)
(28)		MCDWB	4.5	4.2	10.1	19.0	24.6	29.8	30.5	29.0	27.5	22.3	12.8	6.1	(28)	
(29)		2%	WB	1.7	1.2	4.2	9.3	17.5	21.1	23.8	22.2	20.4	15.1	9.1	3.0	(29)
(30)		MCDWB	2.3	2.1	6.8	14.3	21.9	26.2	28.3	26.4	23.8	18.6	10.4	3.7	(30)	
(31)		5%	WB	0.4	0.1	2.8	7.7	14.8	19.8	22.3	21.3	18.8	13.4	7.1	1.7	(31)
(32)		MCDWB	1.0	0.9	4.8	11.5	19.7	23.5	26.2	25.2	21.7	15.9	8.2	2.4	(32)	
(33)		10%	WB	-2.0	-1.1	1.5	6.4	13.2	18.4	21.0	20.3	17.5	11.3	5.8	0.6	(33)
(34)		MCDWB	-1.0	-0.2	3.2	9.3	17.7	22.2	24.8	24.1	20.4	13.2	7.0	1.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.3	7.9	7.7	7.4	8.6	9.0	9.3	9.1	9.1	6.9	5.7	6.2	(35)	
(36)		5% DB	MCDBR	8.0	7.2	9.1	13.0	14.7	13.1	12.1	11.9	12.3	11.4	8.2	6.0	(36)
(37)		MCWBR	8.0	6.2	6.5	7.7	7.6	6.6	6.1	6.0	6.5	7.4	6.6	5.7	(37)	
(38)		5% WB	MCDBR	7.6	6.5	8.2	10.8	12.3	10.8	10.5	9.9	10.0	10.3	7.2	5.8	(38)
(39)		MCWBR	7.8	5.9	6.3	7.2	7.5	6.7	6.1	5.8	6.4	7.1	6.3	5.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.267	0.287	0.329	0.356	0.371	0.407	0.418	0.399	0.374	0.342	0.311	0.295	(40)	
(41)		taud	2.312	2.243	2.208	2.333	2.387	2.285	2.274	2.330	2.393	2.471	2.460	2.274	(41)	
(42)		Ebn at Noon	828	881	888	899	894	859	843	846	836	813	773	733	(42)	
(43)		Edh at Noon	74	98	120	117	115	129	129	117	100	80	66	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.40	2.31	3.51	4.44	5.06	5.51	5.39	4.69	3.68	2.15	1.24	0.94	(44)	
(45)		RadStd	0.19	0.28	0.28	0.38	0.53	0.35	0.53	0.30	0.31	0.18	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page